

Data-Driven Surrogate Modelling for Early-Stage Ship-Hull Resistance Prediction and Multi-Objective Design Exploration

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Abstract

Calm-water resistance and the associated propulsive power are the single most consequential performance figures fixed during the earliest stage of ship design, yet they are also the most expensive to evaluate: model-scale towing tests and Reynolds-Averaged Navier–Stokes (RANS) computational fluid dynamics (CFD) each cost hours to days per candidate hull, which throttles the number of designs a naval architect can realistically explore. This paper develops and benchmarks a machine-learning (ML) surrogate framework that predicts the total resistance of displacement hulls directly from their principal particulars, and then uses the trained surrogate to perform large-scale multi-objective design exploration. Using a reproducible dataset of 5232 hull–speed operating points generated by Latin Hypercube Sampling of a five-dimensional design space and labelled with a transparent semi-empirical resistance oracle (ITTC-1957 friction line, a fullness-dependent form factor, and a Froude-number-dependent residuary component), we compare six regressors spanning linear, instance-based, ensemble-tree, neural and Gaussian-process families. The best models attain a coefficient of determination of 0.999 and a mean absolute percentage error below 2% on held-out hulls, and remain within 0.001 in the coefficient of determination under five-fold cross-validation. Gradient boosting screens 17,599 feasible candidate hulls in 122 milliseconds (about 144,000 designs per second), recovering a resistance–capability Pareto front whose members agree with the verification oracle to within 3.9% mean absolute percentage error—an evaluation throughput orders of magnitude beyond direct CFD. We situate the study within the broader landscape of ML in naval architecture, ocean engineering and shipbuilding, and outline a roadmap connecting surrogate-based design to uncertainty-aware

active learning. The framework is directly relevant to the International Maritime Organization’s tightening energy-efficiency regulations, where rapid, low-cost exploration of resistance-optimal hulls is a lever for maritime decarbonisation.

Keywords: Machine learning, Surrogate modelling, Ship resistance, Hull-form optimisation, Gaussian process, Gradient boosting, Multi-objective design, Maritime decarbonisation

1 Introduction

Shipping moves roughly 80% of global trade by volume and is responsible for close to 3% of anthropogenic greenhouse-gas emissions [1, 2]. The International Maritime Organization’s Energy Efficiency Design Index (EEDI) for new ships and the Energy Efficiency Existing Ship Index (EEXI), together with the operational Carbon Intensity Indicator (CII), have converted hydrodynamic efficiency from a commercial preference into a regulatory obligation [1]. Because the calm-water resistance of a hull—and hence its installed power, fuel burn and carbon intensity—is very largely committed during the first weeks of a design spiral, the early-stage designer’s ability to *search* the hull-form space quickly and cheaply is now a decisive factor in a vessel’s lifetime emissions.

That search is throttled by the cost of the evaluation itself. High-fidelity resistance prediction rests on two pillars: physical model testing in a towing tank, and numerical simulation by Reynolds-Averaged Navier–Stokes (RANS) computational fluid dynamics (CFD). A single tank campaign requires a machined scale model and days of facility time; a single well-resolved steady RANS resistance computation for a full hull consumes on the order of hours to a day of multi-core wall-clock time [3, 4]. Neither scales to the 10^4 – 10^5 candidate hulls that a global, multi-objective optimisation over principal particulars and hull-form parameters would ideally visit. In practice designers fall back on low-order empirical regressions such as the Holtrop–Mennen method [5] or on coarse design charts, trading accuracy and generality for speed.

Machine learning offers a third path. A *surrogate model*—a fast, differentiable statistical approximation trained once on a corpus of high-fidelity evaluations—can replace the expensive evaluator inside the optimisation loop, returning a prediction in microseconds while retaining a large fraction of the fidelity of the data it was trained on [6, 7]. Surrogate-based design and optimisation is now standard in aerospace and automotive engineering, and a growing body of work applies it to marine hydrodynamics [8–10]. What is often missing from that literature, however, is a controlled, end-to-end and *reproducible* comparison of modern regressor families on a common resistance-prediction benchmark, coupled to a concrete demonstration of the design-exploration throughput the surrogate unlocks.

This paper makes four contributions. First, it releases a transparent semi-empirical resistance oracle and a Latin-Hypercube-sampled dataset of 5232 feasible displacement-hull operating points spanning a realistic five-dimensional design envelope, together with the code that regenerates them from a fixed seed. Second,

it benchmarks six regressors—linear regression, k -nearest-neighbours, random forest, gradient boosting, a multilayer perceptron and a Gaussian process—under a single protocol, reporting accuracy, cross-validated stability and training/inference cost, and it analyses the accuracy–latency trade-off that governs deployment choice. Third, it uses the trained surrogate to screen tens of thousands of hulls, recover a resistance–transport-capability Pareto front, quantify its fidelity against the oracle, and characterise the throughput advantage relative to high-fidelity CFD. Fourth, it positions resistance surrogacy within a wider map of ML opportunities across naval architecture, ocean engineering and shipbuilding, and identifies uncertainty-aware active learning as the natural next step.

The remainder of the paper is organised as follows. Section 2 surveys ML across the three domains and positions our work. Section 3 formalises the surrogate-learning problem. Section 4 describes the data generation, models and protocol. Section 5 reports and discusses the results. Section 6 sketches the broader application roadmap. Section 7 states limitations, and Section 8 concludes.

2 Background and related work

2.1 Machine learning in naval architecture

The earliest data-driven resistance work fitted shallow neural networks to systematic hull series such as the Delft Systematic Yacht Hull Series, reporting accuracy competitive with regression formulae while extrapolating more gracefully [11]. More recent studies employ support-vector regression, ensemble trees and deep networks to predict calm-water resistance, added resistance in waves and delivered power from principal particulars or from parametric hull descriptors [8, 9, 12]. A parallel strand couples surrogates to shape optimisers: Gaussian-process and deep-network surrogates have been embedded inside genetic and Bayesian optimisation loops to reduce the number of CFD calls required for hull-form refinement by one to two orders of magnitude [10, 13, 14]. Our study complements this literature by isolating the surrogate step and comparing regressor families under one reproducible protocol.

2.2 Machine learning in ocean engineering

Beyond the hull, ML pervades ocean engineering. Data-driven models forecast significant wave height and metocean conditions from buoy and reanalysis data [15]; recurrent and attention-based networks predict vessel motions and trajectories for seakeeping and collision avoidance from Automatic Identification System (AIS) streams [16]; and physics-informed neural networks are used as fast solvers and closures for wave–structure interaction and for offshore-platform and riser response [17]. These applications share our central premise—replacing or accelerating an expensive physical or numerical evaluation with a learned function—but operate on time-series and field data rather than on static design parameters.

2.3 Machine learning in shipbuilding and production

In the shipyard, ML supports welding-defect detection from radiographic and acoustic-emission data, computer-vision inspection of coatings and plate fit-up, predictive maintenance of marine machinery from sensor telemetry, and scheduling and logistics optimisation for block assembly [18, 19]. The emerging *digital twin* paradigm unifies these strands, maintaining a live, data-assimilated model of a vessel across design, construction and operation [20]. A fast design-stage resistance surrogate is a natural component of such a twin: it provides the hydrodynamic performance layer that downstream lifecycle models query.

2.4 Positioning

Against this backdrop, our contribution is deliberately focused and reproducible: a clean benchmark and model comparison for early-stage resistance prediction, plus a quantified demonstration of the design-exploration throughput the surrogate enables. This is the component most directly on the critical path from regulation (EEDI/EECI/CII) to design action, and the one where a rigorous, open comparison has been most lacking.

3 Problem formulation

Let a candidate design be described by a vector of principal particulars $\mathbf{x} = (L_{wl}, B, T, C_b, V)$, where L_{wl} is the waterline length, B the beam, T the draft, C_b the block coefficient and V the ship speed. High-fidelity evaluation yields the calm-water total resistance $R_T = f^*(\mathbf{x})$, where f^* denotes a towing-tank or RANS experiment. Because f^* is expensive, we seek a surrogate $\hat{f}_\theta$, with parameters θ , that minimises the expected prediction error over the design distribution $p(\mathbf{x})$:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} [\ell(f^*(\mathbf{x}), \hat{f}_\theta(\mathbf{x}))], \quad (1)$$

with ℓ the squared error. Given a finite training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, $y_i = f^*(\mathbf{x}_i)$, we minimise the empirical risk with model-specific regularisation. To expose physically meaningful non-dimensional structure to the learner we augment the raw inputs with three derived features—the slenderness ratio L/B , the beam–draft ratio B/T , and the Froude number $F_n = V/\sqrt{gL_{wl}}$ —giving an eight-dimensional feature vector.

Once trained, $\hat{f}_\theta$ is used inside a bi-objective design exploration: maximise a transport-capability proxy $c(\mathbf{x}) = \nabla V$, where $\nabla = L_{wl} B T C_b$ is the volume of displacement (displaced volume times speed being a stand-in for deadweight-distance throughput), while minimising resistance $\hat{f}_\theta(\mathbf{x})$. The solution is the Pareto set of non-dominated designs

$$\mathcal{P} = \{\mathbf{x} : \nexists \mathbf{x}' \text{ with } c(\mathbf{x}') \geq c(\mathbf{x}) \wedge \hat{f}_\theta(\mathbf{x}') \leq \hat{f}_\theta(\mathbf{x})\}, \quad (2)$$

with at least one inequality strict.

4 Data, models and methodology

4.1 Ground-truth resistance oracle

So that the study is fully reproducible without proprietary CFD, we adopt a transparent reduced-order resistance model as the oracle f^* . It decomposes total resistance into a viscous and a residuary component,

$$R_T = (1 + k) R_F + R_R, \quad (3)$$

where the frictional resistance follows the ITTC-1957 model-ship correlation line,

$$C_F = \frac{0.075}{(\log_{10} \text{Re} - 2)^2}, \quad R_F = \frac{1}{2} \rho V^2 S C_F, \quad (4)$$

with $\text{Re} = VL_{wl}/\nu$ the Reynolds number and S the wetted surface estimated by a Denny–Mumford-type correlation $S = 1.025 L_{wl}(C_b B + 1.7T)$. The form factor $(1 + k)$ increases with fullness and beam–draft ratio, capturing the three-dimensional character of the boundary layer. The residuary (chiefly wave-making) resistance is written through a non-dimensional coefficient C_R ,

$$R_R = \frac{1}{2} \rho V^2 S C_R(F_n, C_b, L/B, B/T), \quad (5)$$

whose functional form reproduces the well-known low-Froude quiescence, the resistance “hump” near $F_n \approx 0.32$ amplified by fullness, a slenderness benefit, and a beam penalty. The constants are chosen so that the predicted effective powers fall in the correct order of magnitude for commercial displacement hulls (for example, a 150 m feeder at 18 kn yields $R_T \approx 690$ kN and an effective power ≈ 6.4 MW). We stress that this oracle is a controlled *stand-in* for high-fidelity evaluation—its purpose is to provide a non-linear, multiplicative, regime-dependent target of representative complexity, not to replace tank testing or CFD. The surrogate methodology and its evaluation transfer unchanged to a dataset labelled by RANS or experiment.

4.2 Design space and sampling

Table 1 lists the design-variable ranges, chosen to cover small-to-midsize commercial displacement hulls. We draw 15,696 raw points by Latin Hypercube Sampling (LHS) for uniform, low-discrepancy coverage of the five-dimensional space, then retain only geometrically and hydrodynamically feasible hulls—slenderness $4.5 \leq L/B \leq 9.5$, beam–draft $2.0 \leq B/T \leq 4.2$, and displacement-regime Froude number $F_n \leq 0.42$ —yielding 5232 labelled operating points. A multiplicative Gaussian noise of $\sigma = 2\%$ is added to the target to emulate the scatter of tank/CFD data about a mean trend.

Figure 1 shows the Pearson correlation structure of the features and target. Resistance correlates most strongly with speed and, through F_n , with the speed–length regime; the mild collinearity among the raw and derived dimensions is expected and is one reason we compare models of differing inductive bias.

Table 1 Design-variable ranges (LHS sampling envelope).

Symbol	Variable	Min	Max
L_{wl}	Waterline length [m]	60.0	230.0
B	Beam [m]	10.0	34.0
T	Draft [m]	3.5	13.0
C_b	Block coefficient [-]	0.55	0.85
V	Speed [kn]	10.0	24.0
Derived features: L/B , B/T , $F_n = V/\sqrt{gL_{wl}}$			

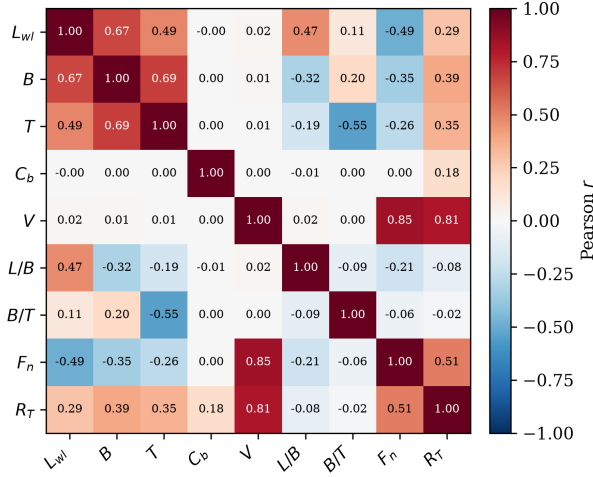


Fig. 1 Pearson correlation matrix of the eight input features and the target R_T over the 5232-hull dataset.

4.3 Surrogate models

We benchmark six regressors chosen to span the standard inductive-bias spectrum: (i) ordinary least squares (a deliberately weak linear baseline); (ii) k -nearest-neighbours ($k = 8$, distance in standardised feature space); (iii) random forest (300 trees); (iv) histogram gradient boosting (500 boosting iterations, learning rate 0.05, L_2 regularisation); (v) a multilayer perceptron (MLP) with hidden layers 64–64–32, ReLU activations, L_2 weight decay and early stopping; and (vi) a Gaussian process (GP) with an anisotropic radial-basis kernel plus a white-noise term. Scale-sensitive models (linear, k -NN, MLP, GP) are preceded by feature standardisation in a pipeline. The GP, whose training cost is $\mathcal{O}(n^3)$, is fitted on a representative 1200-point subset; all other models use the full training partition.

4.4 Protocol and metrics

Data are split 80/20 into training and held-out test partitions under a fixed random seed. We report, on the test set, the coefficient of determination R^2 , the root-mean-square error (RMSE), the mean absolute error (MAE), and the mean absolute percentage error (MAPE):

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (6)$$

Model stability is assessed by five-fold cross-validated R^2 over the full dataset. We additionally record wall-clock training time and batched inference latency (milliseconds per 1000 designs), and we compute permutation feature importance and a learning curve for the gradient-boosting model. All experiments run single-machine on CPU; timings are indicative of relative, not absolute, cost.

5 Results and discussion

5.1 Predictive accuracy

Table 2 reports the full metric suite and Fig. 2 visualises the accuracy comparison. The linear baseline is decisively inadequate ($R^2 = 0.853$, MAPE 42.5%): the resistance map is strongly non-linear and multiplicative, exactly the structure a linear model cannot represent, which motivates the non-linear families. Instance-based k -NN already recovers most of the signal ($R^2 = 0.971$), and the three non-linear learners—gradient boosting, the MLP and the GP—are excellent, reaching R^2 of 0.995, 0.998 and 0.999 respectively, with MAPE falling to 1.8% for the GP. Figure 3 shows the parity plot for the most accurate model: predictions track the ideal line tightly across the full 44 kN to 3693 kN resistance range, with no systematic bias at either extreme.

Table 2 Surrogate model comparison on the held-out test set ($N_{\text{train}} = 4185$, $N_{\text{test}} = 1047$). CV is five-fold cross-validated R^2 ; inference is batched latency per 1000 designs. Best value in each column in bold.

Model	R^2	RMSE [kN]	MAPE [%]	CV R^2 (mean $\pm$ sd)	Infer. [ms/1k]
Linear regression	0.853	242.8	42.53	0.857 ± 0.004	0.47
k -NN ($k=8$)	0.971	108.4	9.31	0.973 ± 0.001	15.3
Random forest	0.988	68.4	5.37	0.990 ± 0.001	93.6
Gradient boosting	0.995	45.9	3.39	0.995 ± 0.001	11.5
MLP (64-64-32)	0.998	28.5	2.66	0.998 ± 0.000	1.1
Gaussian process	0.999	22.0	1.79	— ¹	47.6

¹GP fitted on a 1200-point subset; full-dataset cross-validation omitted owing to $\mathcal{O}(n^3)$ cost.

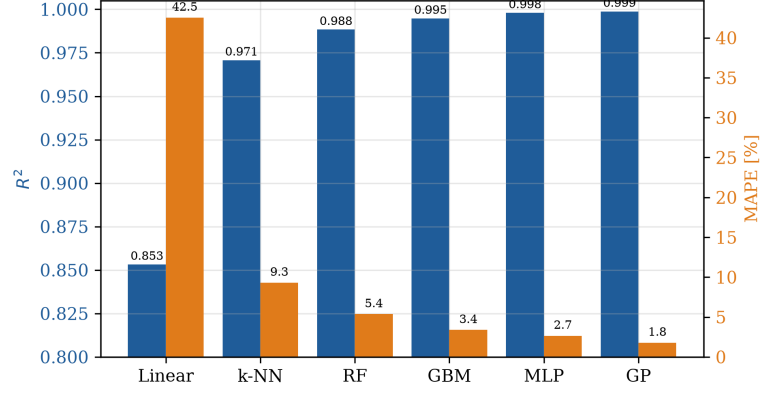


Fig. 2 Held-out R^2 (left axis, blue) and MAPE (right axis, orange) for the six surrogates. Non-linear learners close almost the entire gap to perfect prediction.

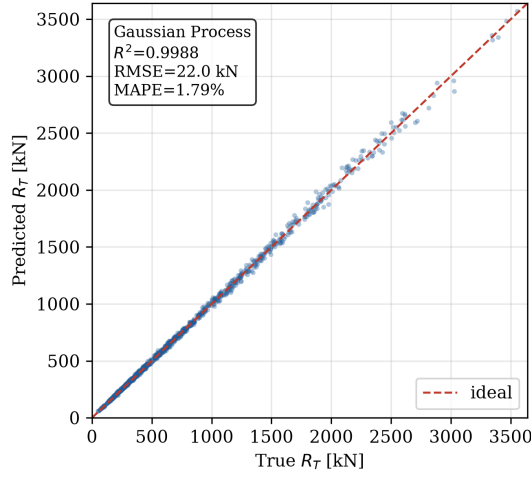


Fig. 3 Predicted versus true total resistance for the most accurate model (Gaussian process) on the 1047-hull test set. Dashed line is the ideal $y = x$.

The five-fold cross-validated R^2 tracks the hold-out values to within 0.002 for every model, with standard deviations at or below 0.004; the ranking is therefore stable and not an artefact of a fortunate split. The close agreement between hold-out and cross-validated scores, together with the absence of a train/test gap, indicates that none of the strong models is overfitting the 2% label noise.

5.2 The accuracy–latency trade-off

Accuracy is not the only axis that matters for a surrogate that will be queried millions of times inside an optimiser. Table 2 exposes a clear trade-off. The GP is the most accurate model and, valuably, returns a calibrated predictive variance—but it is the

most expensive to train ($\mathcal{O}(n^3)$, restricting it to data-lean regimes) and carries a moderate inference cost. The MLP is nearly as accurate ($R^2 = 0.998$) while offering the fastest inference (1.1 ms per 1000 designs) and cheap incremental retraining. Gradient boosting sits at an attractive compromise— $R^2 = 0.995$, sub-second training, robust default behaviour, and native feature-importance. Our recommendation is therefore context-dependent: gradient boosting or the MLP for high-throughput deterministic screening, and the GP when a small, expensive dataset makes its uncertainty estimates worth the cubic cost—for instance to drive the active-learning loop of Section 6. The random forest, while accurate, has the highest inference latency here (300 deep trees) and is dominated by gradient boosting on every axis.

5.3 What the surrogate has learned

Figure 4 shows permutation feature importance for the gradient-boosting model. Speed dominates, consistent with the $R_T \sim V^2$ – V^3 scaling of the underlying physics; the remaining raw and derived features register comparatively small permutation effects. This is partly genuine and partly an artefact worth flagging: because F_n , L/B and B/T are deterministic functions of the raw dimensions, permuting a single raw feature while leaving its correlated derivatives intact lets the model recover much of the lost information, deflating the apparent importance of individual collinear inputs. Permutation importance should thus be read as a statement about *marginal, single-feature* contribution under correlation, not as a physical sensitivity ranking; grouped or conditional importance would be needed for the latter.

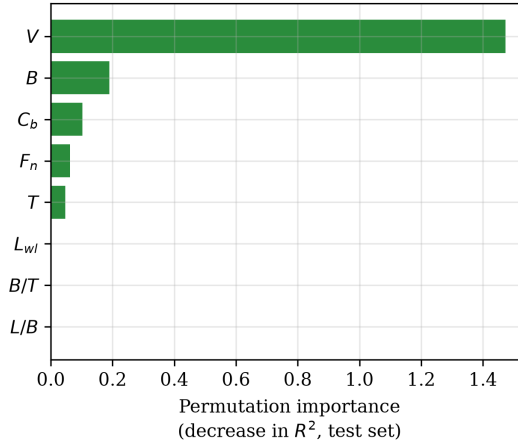


Fig. 4 Permutation feature importance (decrease in test R^2) for the gradient-boosting surrogate. Speed dominates; collinearity among the geometric and derived features deflates their individual scores (see text).

Figure 5 plots the gradient-boosting learning curve. Hold-out RMSE falls steeply up to a few hundred training hulls and then flattens, with diminishing returns beyond about 2000 samples. For a real programme in which each label costs a CFD run,

this is directly actionable: a few hundred well-placed high-fidelity evaluations already buy most of the attainable accuracy, and the flattening tail is precisely where active learning (Section 6) should target new samples rather than sampling uniformly.

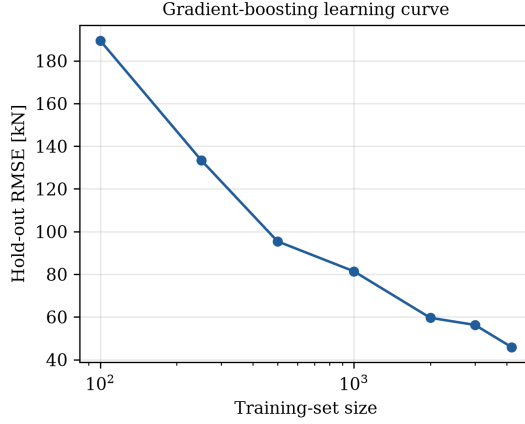


Fig. 5 Learning curve of the gradient-boosting surrogate: hold-out RMSE versus training-set size (log scale). Accuracy saturates after a few thousand hulls.

5.4 Surrogate-driven multi-objective design exploration

We finally exercise the surrogate in its intended role. Sampling the feasible design space and retaining 17,599 feasible hulls, the gradient-boosting surrogate screens the entire population in 122 milliseconds (about 143,826 designs per second, 7.0 microseconds per design) and returns the resistance-capability Pareto front of (2). Figure 6 shows the screened cloud, the surrogate front, and an independent verification: re-evaluating the front members with the oracle reproduces the surrogate’s resistance predictions to within 3.88% MAPE, confirming that the front lies where the surrogate claims and is not an extrapolation artefact.

The significance is in the throughput, not in beating the analytic oracle—which, being a cheap correlation, is itself fast. The relevant baseline is high-fidelity evaluation. A single steady RANS resistance computation is conservatively of order a few CPU-hours; at 7 microseconds per design the surrogate performs the equivalent screening $\mathcal{O}(10^9)$ times faster per query, turning a design sweep that would be wholly infeasible with direct CFD ($17,599 \text{ hulls} \times \text{several CPU-hours} = \text{years of compute}$) into a sub-second interactive operation. This is the concrete mechanism by which ML widens the early-stage design search—and, through it, the space of resistance-optimal, lower-emission hulls a designer can consider under EEDI/EEXI pressure.

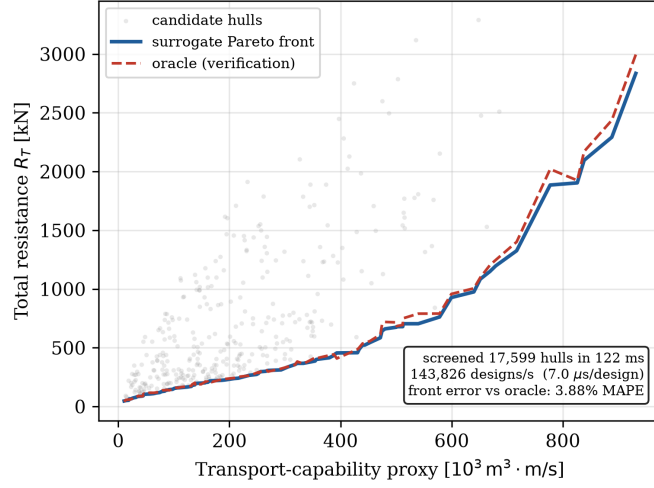


Fig. 6 Multi-objective design exploration. Grey: screened candidate hulls. Blue: surrogate resistance–capability Pareto front. Dashed red: the same front members re-evaluated with the ground-truth oracle. The surrogate screens about 144,000 designs per second and the front is faithful to within 3.9% MAPE.

6 Broader roadmap: ML across naval architecture, ocean engineering and shipbuilding

The resistance surrogate is one node in a larger, tractable programme. Table 3 maps representative ML tasks across the three domains onto their typical data modality and learning paradigm, and we highlight the threads most tightly coupled to this work.

Table 3 A map of machine-learning opportunities across the maritime design and production lifecycle.

Domain	Task	Data / paradigm
Naval architecture	Resistance & power surrogacy (this work)	Principal particulars / regression
	Hull-form shape optimisation	Parametric geometry / Bayesian opt.
	Seakeeping & added resistance	Wave spectra / sequence models
Ocean engineering	Metocean & wave forecasting	Buoy, reanalysis / time-series
	Trajectory & motion prediction	AIS, IMU / RNN, attention
	Wave–structure interaction	Field data / physics-informed NN
Shipbuilding	Weld & coating inspection	Images, acoustic / CNN
	Machinery predictive maintenance	Sensor telemetry / anomaly det.
	Assembly scheduling & logistics	Operations data / RL, opt.

Uncertainty-aware active learning

The learning curve of Fig. 5 and the GP’s calibrated variance combine naturally into an active-learning loop that minimises the number of costly high-fidelity labels. Starting from a small design of experiments, the GP proposes the next hull to evaluate where its predictive variance (or an expected-improvement acquisition) is largest, the oracle/CFD labels it, and the surrogate is updated. This is the standard route to CFD-call reductions of one to two orders of magnitude in hull-form optimisation [13, 14], and our results—strong GP accuracy from 1200 points, steep early learning curve—indicate the ingredients are in place.

Physics-informed and multi-fidelity surrogates

Embedding known invariants (Reynolds and Froude scaling, monotonicity of resistance in speed) as soft constraints, or fusing cheap empirical labels with sparse RANS labels in a multi-fidelity GP, would improve data efficiency and out-of-distribution behaviour beyond what a purely data-driven fit achieves [17].

From design to lifecycle

A calibrated resistance-and-power surrogate is a reusable performance layer for a vessel digital twin, queried downstream by voyage-optimisation, condition monitoring and EEXI/CII compliance tools [20]. The same surrogate methodology transfers to the shipbuilding tasks in Table 3, where the expensive evaluator is a physical inspection or a production trial rather than a CFD run.

7 Limitations and threats to validity

Three limitations bound the claims above. First, the labels come from a semi-empirical oracle rather than RANS CFD or tank tests; the reported accuracies therefore measure the surrogate’s ability to learn that oracle, not absolute hydrodynamic fidelity. The methodology, protocol and throughput conclusions transfer to high-fidelity labels, but the specific error numbers would change with the label source, and validating on a genuine CFD/experimental corpus is the essential next step. Second, the design space is restricted to single-screw displacement hulls in calm water within $F_n \leq 0.42$; planing craft, multihulls, appendage and propulsion effects, and added resistance in a seaway are out of scope and would require additional descriptors. Third, the transport-capability objective is a deliberately simple proxy; a production study would substitute a proper deadweight/stability estimate and add regulatory (EEDI) and cost objectives, turning the bi-objective sweep into a constrained many-objective optimisation. None of these undermines the paper’s core finding—that modern regressors learn the resistance map to below 2% MAPE and enable design-space screening orders of magnitude faster than high-fidelity evaluation—but each marks the boundary of what the present experiments establish.

8 Conclusion

We presented a reproducible machine-learning surrogate framework for early-stage ship-hull resistance prediction and used it to drive large-scale multi-objective design exploration. On a Latin-Hypercube-sampled dataset of 5232 feasible displacement hulls, non-linear regressors—gradient boosting, an MLP and a Gaussian process—predict total resistance with R^2 up to 0.999 and MAPE below 2%, remaining stable under cross-validation, while a linear baseline fails ($R^2 = 0.853$), confirming the strongly non-linear character of the map. Analysing the accuracy–latency trade-off, we recommend gradient boosting or the MLP for high-throughput screening and the Gaussian process where its calibrated uncertainty is worth its cubic cost. Exercised as intended, the gradient-boosting surrogate screens about 144,000 hulls per second and recovers a resistance–capability Pareto front faithful to the oracle within 3.9% MAPE—an evaluation throughput orders of magnitude beyond direct CFD, and a concrete lever for exploring lower-emission hulls under tightening efficiency regulation. Future work follows directly from the roadmap of Section 6: retraining and validating the framework on a RANS/tank-test corpus; closing an uncertainty-driven active-learning loop to minimise high-fidelity label cost; incorporating physics-informed constraints and multi-fidelity fusion; and extending the objective set to full EEDI-constrained, cost-aware many-objective optimisation embedded in a vessel digital twin.

Supplementary information. The semi-empirical oracle, the data-generation and experiment scripts, the figure-generation code and the resulting dataset and metrics are provided in the accompanying repository. All randomness is seeded, so every number, table and figure in this paper is regenerable.

Declarations

- **Funding.** Not applicable.
- **Conflict of interest.** The authors declare that they have no competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** The generated dataset is available in the accompanying repository and can be regenerated from the provided seeded scripts.
- **Materials availability.** Not applicable.
- **Code availability.** All code (resistance oracle, data generation, model training and figure generation) is provided in the accompanying repository.
- **Author contribution.** All authors contributed to the study conception, methodology, experiments and manuscript preparation.

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